

Mathematics of Machine Learning

Probability

$$Y = g(X)$$

 Ω

$y_{13} = g(x_{13})$	$y_{14} = g(x_{14})$	$y_{15} = g(x_{15})$	$y_{16} = g(x_{16})$
$y_9 = g(x_9)$	$y_{10} = g(x_{10})$	$y_{11} = g(x_{11})$	$y_{12} = g(x_{12})$
$y_5 = g(x_5)$	$y_6 = g(x_6)$	$y_7 = g(x_7)$	$y_8 = g(x_8)$
$y_1 = g(x_1)$	$y_2 = g(x_2)$	$y_3 = g(x_3)$	$y_4 = g(x_4)$

Probability, Random Variables, Expectations

Outline

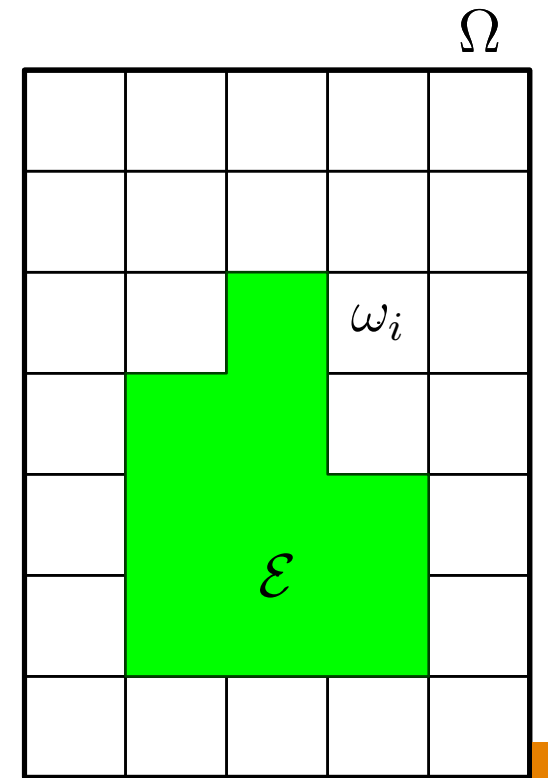
1. **Random Variables**
2. Expectations
3. Calculus of Probabilities

$$\Omega$$

x_{31}	x_{32}	x_{33}	x_{34}	x_{35}
x_{26}	x_{27}	x_{28}	x_{29}	x_{30}
x_{21}	x_{22}	x_{23}	x_{24}	x_{25}
x_{16}	x_{17}	x_{18}	x_{19}	x_{20}
x_{11}	x_{12}	x_{13}	x_{14}	x_{15}
x_6	x_7	x_8	x_9	x_{10}
x_1	x_2	x_3	x_4	x_5

Modelling Uncertainty

- To model a world with uncertainty we consider some set of **elementary events** or **outcomes** Ω ■
- For the outcome of rolling a dice $\Omega = \{1,2,3,4,5,6\}$ ■
- The elementary events ω_i are **mutually exclusive** $\omega_i \cap \omega_j = \emptyset$ and **exhaustive** $\bigcup_i \omega_i = \Omega$ ■
- We consider **events** $\mathcal{E} = \bigcup_{i \in \mathcal{I}} \omega_i$ ■
- E.g. For a dice throw $\mathcal{E} = \{2,4,6\}$ ■



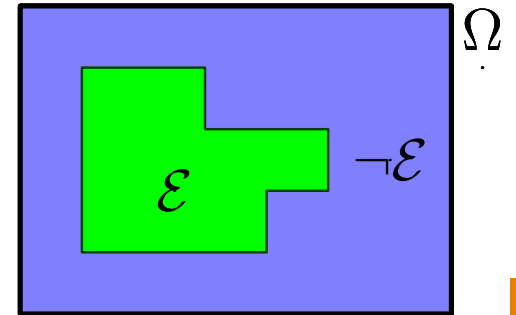
Probabilities

- We attribute a **probability**, $\mathbb{P}(\mathcal{E})$, to an event, $\mathcal{E}$, with the requirements

- ★ $0 \leq \mathbb{P}(\mathcal{E}) \leq 1$ ■

- ★ $\mathbb{P}(\mathcal{E}) + \mathbb{P}(\neg\mathcal{E}) = 1$ where $\neg\mathcal{E} = \Omega \setminus \mathcal{E}$

■



■

- In some cases we can interpret $\mathbb{P}(\mathcal{E})$ as the expected frequency of occurrence of a repetitive trial ■
- But $\mathbb{P}(\text{Pass COMP6208 exam})$ is something you do once ■
- Can think of probability as an informed belief that something might happen ■
- When our knowledge changes the probability changes ■

Random Variables

- We can define a **random variable**, X , by partition the set of outcomes Ω and assign a numbers to each partition■
- E.g. for a dice

$$X = \begin{cases} 0 & \text{if } \omega \in \{1,3,5\} \\ 1 & \text{if } \omega \in \{2,4,6\} \end{cases}$$

- $\mathbb{P}(X = x_i) = \mathbb{P}(\mathcal{E}_i)$ where $\mathcal{E}_i$ is the event that corresponding to the partition with value x_i ■

				Ω
	x_1		x_4	
				x_5
		x_3		
x_2				

What's In A Name

- We denote random variables with capital letters, X , Y , Z , etc.■
- The symbol denote an object that can take one of a number of different values, but which one is still to be decided by chance■
- When we write $\mathbb{P}(X)$ we can view this as short-hand for

$$(\mathbb{P}(X = x) \mid x \in \mathcal{X}) = (\mathbb{P}(X = x_1), \mathbb{P}(X = x_2), \dots, \mathbb{P}(X = x_n))$$

where $\mathcal{X}$ is the set of possible values that X can take■

- We treat random variables very differently to normal numbers (scalars) when we consider taking expectations■

Function of Random Variables

- Any function, $Y = g(X)$, of a random variable, X , is a random variable

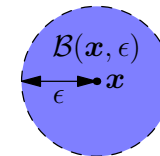
$$Y = g(X) \quad \Omega$$

$y_{13} = g(x_{13})$	$y_{14} = g(x_{14})$	$y_{15} = g(x_{15})$	$y_{16} = g(x_{16})$
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Continuous Spaces

- If the space of elementary events is continuous (e.g. for darts $\mathbf{x} = (x, y)$) then $\mathbb{P}(\mathbf{X} = \mathbf{x}) = 0$ ■
- But if we consider a region, $\mathcal{R}$, then we can assign a probability to landing in the region $\mathbb{P}(\mathbf{X} \in \mathcal{R})$ ■
- It is useful to work with **probability densities function** (PDF)

$$f_{\mathbf{X}}(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{\mathbb{P}(\mathbf{X} \in \mathcal{B}(\mathbf{x}, \epsilon))}{\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))}$$



where $\mathcal{B}(\mathbf{x}, \epsilon)$ is a ball of radius ϵ around the point $\mathbf{x}$ and $\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))$ is the volume of the ball■

- If we make a change of variable the volume $\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))$ might change so $f_{\mathbf{X}}(\mathbf{x})$ will change■

Change of Variables

- Consider a region $\mathcal{R}$ —we can describe this using different coordinate systems x or $y = g(x)$ ■

- But

$$\mathbb{P}(X \in \mathcal{R}) = \int_{\mathcal{R}} f_X(x) dx = \mathbb{P}(Y \in \mathcal{R}) = \int_{\mathcal{R}} f_Y(y) dy \blacksquare$$

- As this is true for any region $\mathcal{R}$: $f_X(x) |dx| = f_Y(y) |dy|$ ■

- Or

$$f_X(x) = f_Y(y) \left| \frac{dy}{dx} \right| = f_Y(g(x)) |g'(x)| \blacksquare$$

- A probability density measured in units of probability per cm is different to that measured in units of probability per inch■

Jacobian

- In high dimension if we make a change of variables $\mathbf{x} \rightarrow \mathbf{y}(\mathbf{x})$ (which can be seen as a change of random variables $\mathbf{X} \rightarrow \mathbf{Y}(\mathbf{X})$)
- Then

$$f_{\mathbf{X}}(\mathbf{x}) = f_{\mathbf{Y}}(\mathbf{y}) |\det(\mathbf{J})|$$

where $\mathbf{J}$ is the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \cdots & \frac{\partial y_n}{\partial x_n} \end{pmatrix}$$

- Ensures integrals over volumes are the same

Meaning of Probability Densities

- Probability densities are not probabilities■
- They are positive, but don't need to be less than 1■
- Note that

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x \leq X < x + \delta x)}{\delta x} \blacksquare$$

- We can think of $f_X(x)\delta x$ as $\mathbb{P}(x \leq X < x + \delta x)$ ■
- Note that $f_X(x)\delta x \leq 1$ ■

Cumulative Distribution Functions

- We can define the **cumulative distribution function** (CDF)

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} \sum_{i: x_i \leq x} \mathbb{P}(X = x_i) \\ \int_{-\infty}^x f_X(y) dy \end{cases}$$

- This is a function that goes from 0 to 1 as x goes from $-\infty$ to ∞
- We note that for continuous random variables

$$f_X(x) = \frac{dF_X(x)}{dx}$$

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Expectation

- We can define the expectation of $Y = g(\mathbf{X})$ as

$$\mathbb{E}_{\mathbf{X}}[g(\mathbf{X})] = \begin{cases} \sum_{\mathbf{x} \in \mathcal{X}} g(\mathbf{x}) \mathbb{P}(\mathbf{X} = \mathbf{x}) \\ \int g(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} \end{cases}$$

- The expectation of a constant c is

$$\mathbb{E}_{\mathbf{X}}[c] = \begin{cases} \sum_{\mathbf{x} \in \mathcal{X}} c \mathbb{P}(\mathbf{X} = \mathbf{x}) = c \sum_{\mathbf{x} \in \mathcal{X}} \mathbb{P}(\mathbf{X} = \mathbf{x}) = c \\ \int c f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} = c \int f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} = c \end{cases}$$

- Note $\mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{X}}[g(\mathbf{X})]] = \mathbb{E}_{\mathbf{X}}[g(\mathbf{X})]$

Linearity of Expectation

- Because sums and integrals are linear operators

$$\sum_i (ax_i + by_i) = a \left(\sum_i x_i \right) + b \left(\sum_i y_i \right)$$

$$\int (af(\mathbf{x}) + bg(\mathbf{x}))d\mathbf{x} = a \left(\int f(\mathbf{x})d\mathbf{x} \right) + b \left(\int g(\mathbf{x})d\mathbf{x} \right)$$

then expectations are linear

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y] \blacksquare$$

- Beware usually $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ (unless X and Y are independent) $\blacksquare$

Indicator Functions

- An indicator function has the property

$$\llbracket predicate \rrbracket = \begin{cases} 1 & \text{if } predicate \text{ is True} \\ 0 & \text{if } predicate \text{ is False} \end{cases}$$

(sometimes written $\mathbf{I}_A(x)$ where $A(x)$ is the predicate)■

- We can obtain probabilities from expectations

$$\mathbb{P}(predicate) = \mathbb{E}[\llbracket predicate \rrbracket]■$$

- E.g. The CDF is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{E}[\llbracket X \leq x \rrbracket]■$$

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Joint Probabilities

- Often we want to model complex processes where we have multiple random variables■
- We can define the joint probability

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$$

i.e. the probability of the event where both $X = x$ and $Y = y$ ■

- Clearly $\mathbb{P}(X,Y) = \mathbb{P}(Y,X)$ ■

Marginalisation

- Probabilities are extremely easy to manipulate (although lots of people struggle)■
- One of the most useful properties is known as **marginalisation**

$$\mathbb{P}(X) = \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y)$$

where $\mathcal{Y}$ is the set of values that the random variable Y takes■

- Note that when we write $\mathbb{P}(X)$ we are saying this is true for all values that X can take■
- Although obvious and easy this is extremely useful■

Conditional Probability

- We can also define the probability of an event X given that $Y = y$ has occurred

$$\mathbb{P}(X \mid Y = y) = \frac{\mathbb{P}(X, Y = y)}{\mathbb{P}(Y = y)}$$

- In constructing a model it is often much easier to specify conditional probabilities (because you know something) rather than joint probabilities■
- When manipulating probabilities it is often easier to work with joint probabilities because we can simplify them by marginalising out random variables we are not interested in■

Basic Calculus

- To obtain the joint probability we can use

$$\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y) = \mathbb{P}(Y|X)\mathbb{P}(X) \blacksquare$$

- This generalises to more random variables

$$\mathbb{P}(X,Y,Z) = \mathbb{P}(X,Y|Z)\mathbb{P}(Z) = \mathbb{P}(X|Y,Z)\mathbb{P}(Y|Z)\mathbb{P}(Z) \blacksquare$$

- We can do this in a number of different ways

$$\mathbb{P}(X,Y,Z) = \mathbb{P}(Y,Z|X)\mathbb{P}(X) = \mathbb{P}(Z|Y,X)\mathbb{P}(Y|X)\mathbb{P}(X) \blacksquare$$

- Note that $\mathbb{P}(A,B | X,Y)$ means the probability of random variables A and B given that X and Y take particular values $\blacksquare$

Beware

- Conditional probabilities, $\mathbb{P}(X \mid Y)$ are probabilities for X , but not Y

$$\sum_{x \in \mathcal{X}} \mathbb{P}(X = x \mid Y) = 1 \blacksquare$$

$$\sum_{y \in \mathcal{Y}} \mathbb{P}(X \mid Y = y) \neq 1$$

(in general) $\blacksquare$

- Note that

$$\begin{aligned} \mathbb{E}_Y[\mathbb{P}(X \mid Y)] &= \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y) \mathbb{P}(X \mid Y = y) \blacksquare \\ &= \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y) \blacksquare = \mathbb{P}(X) \blacksquare \end{aligned}$$

Causality

- Conditional probabilities does not imply causality■
- We might have causal relationships

$$\mathbb{P}(\text{pass} \mid \text{study}) = 0.9 \qquad \mathbb{P}(\text{pass} \mid \neg\text{study}) = 0.2■$$

- But if we know $\mathbb{P}(\text{study}) = 0.8$ then we can compute

$$\mathbb{P}(\text{pass}, \text{study}) = \mathbb{P}(\text{pass} \mid \text{study})\mathbb{P}(\text{study}) = 0.9 \times 0.8 = 0.72$$

$$\mathbb{P}(\text{pass}, \neg\text{study}) = \mathbb{P}(\text{pass} \mid \neg\text{study})\mathbb{P}(\neg\text{study}) = 0.2 \times 0.2 = 0.04■$$

and

$$\begin{aligned} \mathbb{P}(\text{study} \mid \text{pass}) &= \frac{\mathbb{P}(\text{pass}, \text{study})}{\mathbb{P}(\text{pass})} \\ &= \frac{\mathbb{P}(\text{pass}, \text{study})}{\mathbb{P}(\text{pass}, \text{study}) + \mathbb{P}(\text{pass}, \neg\text{study})} = \frac{0.72}{0.72 + 0.04} \approx 0.947■ \end{aligned}$$

Independence

- Random variables X and Y are said to be **independent** if

$$\mathbb{P}(X,Y) = \mathbb{P}(X)\mathbb{P}(Y) \blacksquare$$

- Because $\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y)$ and $\mathbb{P}(X,Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$ independence implies

$$\mathbb{P}(X|Y) = \mathbb{P}(X)$$

$$\mathbb{P}(Y|X) = \mathbb{P}(Y) \blacksquare$$

- Probabilistic independence implies a mathematical co-incident not necessarily causal independence $\blacksquare$
- However causal independence implies probabilistic independence $\blacksquare$
- If $X \in \{0,1\}$ represents the outcome of tossing a coin and $Y \in \{1,2,3,4,5,6\}$ the outcome of rolling a dice then X and Y are independent $\blacksquare$

Well Conducted Experiments

- In well conducted experiments we expect the results we obtain are independent■
- Let $\mathcal{D} = (X_1, X_2, \dots, X_m)$ represents possible outcomes from a set of m well conducted experiments then

$$\mathbb{P}(\mathcal{D}) = \prod_{i=1}^m \mathbb{P}(X_i) \blacksquare$$

- Denoting a possible sentence I might say by $\mathcal{S} = (W_1, W_2, \dots, W_m)$ then

$$\mathbb{P}(\mathcal{S}) \neq \prod_{i=1}^m \mathbb{P}(W_i) \blacksquare$$

otherwise it's time I retired■

Conditional Independence

- Let $K(d)$ be a random variable measuring the amount you know about ML on day d of your revision■
- From your revision schedule you can write down your belief

$$\mathbb{P}(K(d) \mid K(d-1), K(d-2), \dots, K(1))■$$

- But a very reasonable model is

$$\mathbb{P}(K(d) \mid K(d-1), K(d-2), \dots, K(1)) = \mathbb{P}(K(d) \mid K(d-1))$$

what you are going to know today will just depend on what you knew yesterday■

- We say that $K(d)$ is **conditionally independent** on $K(d-2)$, $K(d-3)$, etc. given $K(d-1)$ ■

Conclusion

- To work with probabilities you need to know
 - ★ How to go back and forward between joint probabilities and conditional probabilities
 - ★ How to marginalise out variables■
- You need to understand that for continuous outcomes, it makes sense to talk about the probability density■
- You need to know that expectations are linear operators and the expectation of a constant is the constant■
- You need to understand independence■